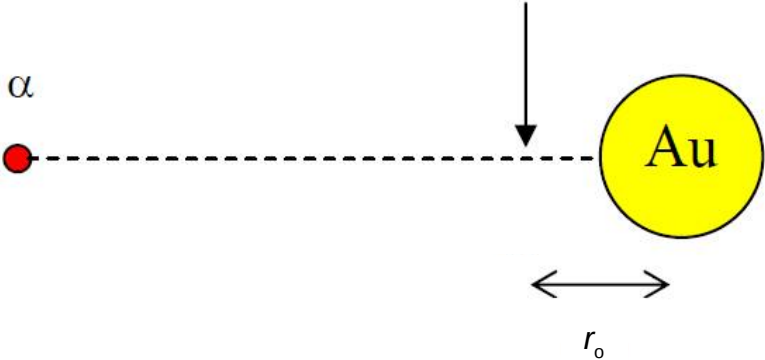
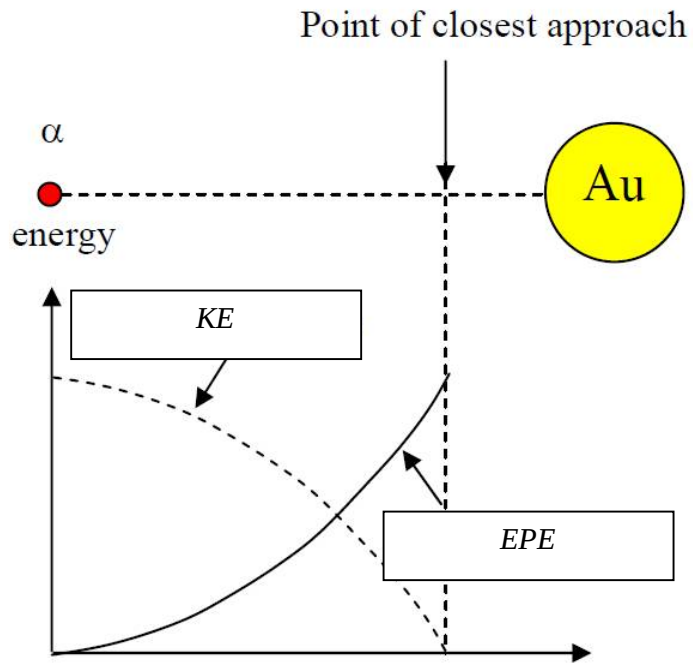


5	(a)	Both the alpha particle and the gold nucleus are positively charged. When the alpha particle travels head-on towards the gold nucleus, all the kinetic energy of the alpha particle is converted to the electric potential energy of the system of the alpha particle and gold nucleus. OR the repulsive force between the alpha particle and the gold nucleus causes deceleration in the alpha particle until it is at its closest approach where it is instantaneously at rest.	B1 B1
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	(b)(i)	Point of closest approach 	
		Loss in K.E. = gain in E.P.E. $5.75(10^6)(1.6 \times 10^{-19}) = \frac{q_\alpha q_{Au}}{4\pi\epsilon_0 r_0}$ $\rightarrow r_0 = \frac{(9.0 \times 10^9)(2e)(79e)}{(9.2 \times 10^{-13})} \approx 4.0 \times 10^{-14} \text{ m}$	C1 A1
	(b)(ii)	$F = \frac{q_\alpha q_{Au}}{4\pi\epsilon_0 r_0^2} = \frac{(9.0 \times 10^9)(2e)(79e)}{(3.96 \times 10^{-14})^2} = 23.3 \text{ N}$	A1
	(c)1./2.	The $^{197}_{79}\text{Au}$ nucleus remains stationary during the interaction between $^{197}_{79}\text{Au}$ and the α-particle. $^{197}_{79}\text{Au}$ and the α-particle are very far apart initially. $^{197}_{79}\text{Au}$ and the α-particle are point charges. The system consists of $^{197}_{79}\text{Au}$ and the α-particle only. The radium nucleus and its daughter nuclide is not part of the system. (Thus, conservation of energy applies for the defined system and the radium nucleus and its daughter nuclide cannot provide an external force on the system.) The gravitational force between the $^{197}_{79}\text{Au}$ and the α-particle is negligible compared to the electric force between them. Any TWO	A1 A1

(d)



A1 -
KE

A1 -
EPE